Algebra 1
Unit 13
Lesson 2 Practice

Name_Answer Key________
Date___________________
Period_________________

Juan deposited $\$20{,}000$ in a savings account with a 4.75% APR compounded quarterly.

1. Write an equation that models the amount of money Juan has in the account.

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4t}$$

2. What is the balance in the account after 8 years?

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4\cdot 8}$$

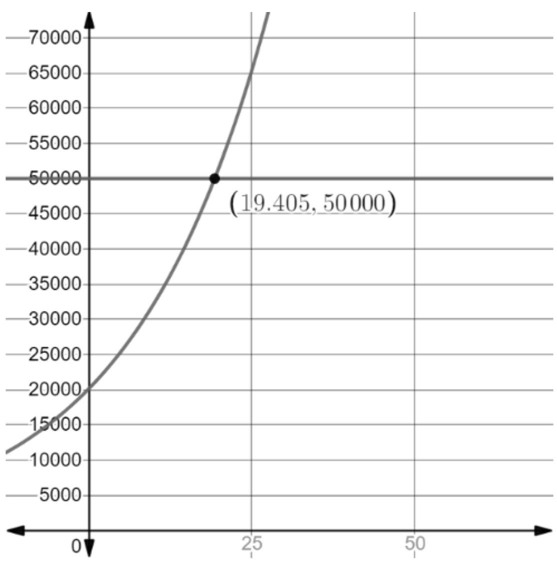
$$= \$29{,}180.30$$

3. What is the balance in the account after 12 years?

$$A = 20{,}000\left(1 + \frac{.0475}{4}\right)^{4\cdot 12}$$

$$A = \$35{,}246.79$$

4. Use the graph to determine when the savings account will reach $\$50{,}000$.



After 19.4 years

For questions 5-10, complete the table below.

	Simple Interest	Compound Interest
Investment amount and APR	$150,000 is invested at an interest rate of 5.25% APR	$150,000 is invested at an interest rate of 5.25% APR compounded monthly
Write the equation.	5. $A = P(1 + rt)$ $A = 150{,}000(1 + .0525t)$	6. $A = P\left(1 + \frac{r}{n}\right)^{nt}$ $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12t}$
Determine the amount after 5 years.	7. $A = 150{,}000(1 + .0525 \cdot 5)$ $A = \$189{,}375$	8. $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12 \cdot 5}$ $A = \$194{,}914.84$
Determine the amount after 25 years.	9. $A = 150{,}000(1 + .0525 \cdot 25)$ $= \$346{,}875$	10. $A = 150{,}000\left(1 + \frac{.0525}{12}\right)^{12 \cdot 25}$ $A = \$555{,}724.43$